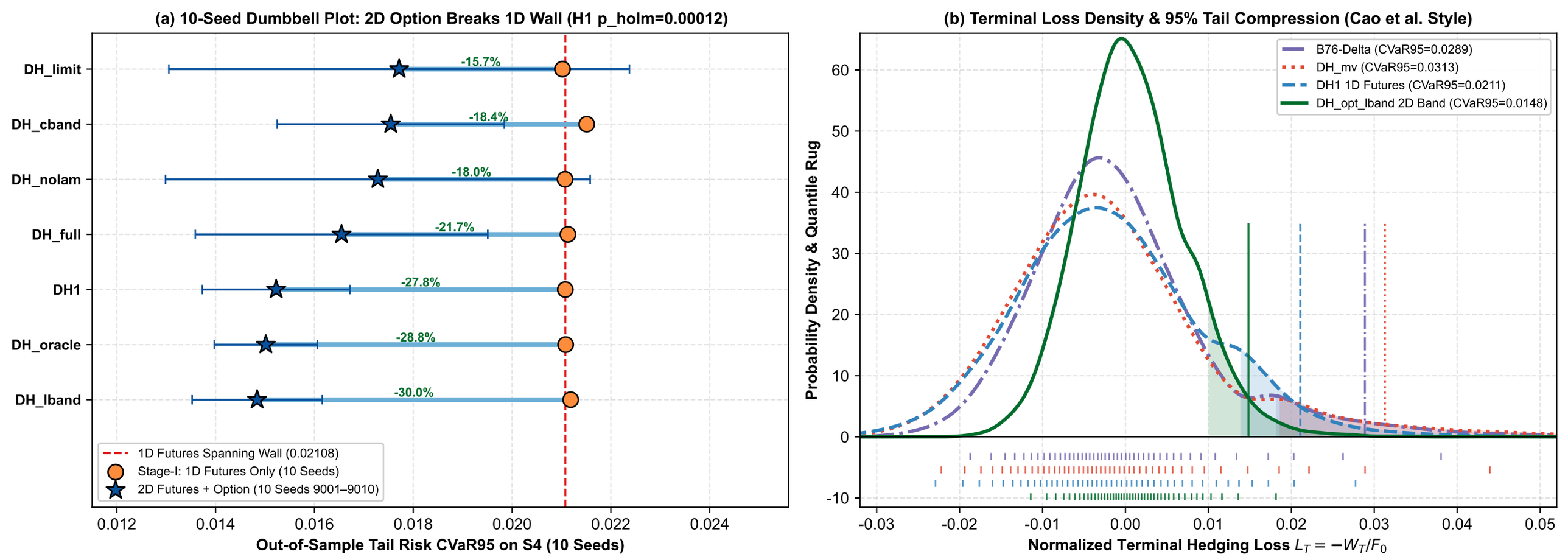

(a) 10-Seed Dumbbell Plot: 2D Option Breaks 1D Wall (H1 p_holm=0.00012)
DH_limit
DH_cband
DH_nolam
DH_full
DH1
DH_oracle
DH_lband
-15.7%
-18.4%
-18.0%
-21.7%
-27.8%
-28.8%
-30.0%
1D Futures Spanning Wall (0.02108)
Stage-I: 1D Futures Only (10 Seeds)
2D Futures + Option (10 Seeds 9001–9010)
0.012
0.014
0.016
0.018
0.020
0.022
0.024
Out-of-Sample Tail Risk CVaR95 on S4 (10 Seeds)
(b) Terminal Loss Density & 95% Tail Compression (Cao et al. Style)
B76-Delta (CVaR95=0.0289)
DH_mv (CVaR95=0.0313)
DH1 1D Futures (CVaR95=0.0211)
DH_opt_lband 2D Band (CVaR95=0.0148)
Probability Density & Quantile Rug
60
50
40
30
20
10
0
-10
-0.03
-0.02
-0.01
0.00
0.01
0.02
0.03
0.04
0.05
Normalized Terminal Hedging Loss $L_T = -W_T/F_0$